CAIRN: A causally-factored, self-certifying world model for interventional generalization

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Abstract

State-of-the-art world models learn a monolithic transition map in which every latent variable depends on every other and actions condition the entire map at once. Because of this structural commitment, a change to a single mechanism of the environment—a damaged actuator, an altered friction coefficient, a failing financial protocol—renders the whole learned model off-support: the model can neither localize the change nor adapt to it, and it fails silently. We introduce CAIRN (Causal Action-conditioned Interventional Rollout Network), a world-model architecture and training algorithm in which (i) latent dynamics factor over a learned sparse causal graph, (ii) actions and exogenous shocks enter as interventions on specific mechanisms in the sense of Pearl's do-operator, implemented architecturally as graph surgery, and (iii) every mechanism carries an internal betting martingale—the *e-gate*—so that the model maintains anytime-valid statistical evidence about which parts of itself are currently wrong. The e-gate drives three capabilities that a monolithic model cannot offer: evidence-triggered localized adaptation that refits only the broken mechanism, evidence-proportional uncertainty inflation that widens predictive intervals for exactly the variables a stale mechanism generates, and online repair of the causal graph itself. We prove that each gate controls its false-alarm probability uniformly over time under a composite null that absorbs the quantile approximation error of learned mechanisms, and we bound its detection delay. On a ground-truth causal benchmark, CAIRN detects unannounced single-mechanism shifts within a median of seventeen steps at a measured false-alarm rate of zero across three independent benchmark seeds, localizes the

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shifted mechanism with an F_1 of 0.90 ± 0.07 where an oracle-tuned CUSUM reaches 0.69 ± 0.12 and ensemble disagreement 0.29 ± 0.06, adapts without any interference on healthy mechanisms where full fine-tuning catastrophically degrades them, and keeps rollout intervals calibrated before, during, and after the shift.

Keywords: world models, causal representation learning, anytime-valid inference, e-processes, distribution shift, model-based reinforcement learning

1. Introduction

Consider a manipulation robot that has spent weeks learning an accurate internal model of its workspace, and suppose that one morning a bearing in its wrist begins to fail, so that the same motor command now produces two thirds of the accustomed torque. Nothing announces this change. The robot's world model—the learned simulator it plans with—continues to imagine futures under the old physics, its planner continues to select actions that would have been excellent yesterday, and the resulting behaviour degrades in ways that are difficult to diagnose precisely because the model reports no more uncertainty today than it did before the bearing failed. The same story plays out wherever learned dynamics models are deployed against a world that can change one piece at a time: a warehouse floor that becomes slippery in one aisle, a chemical process whose catalyst ages, a financial network in which a single protocol's collateral rule is exploited. What all these situations share is that the change is *local*—one mechanism among many—while the failure of the learned model is *global and silent*. This paper is about building world models for which that sentence is no longer true.

A world model is a learned simulator that an agent carries inside itself: given the current state and a candidate course of action, it imagines how the future unfolds, and the agent plans against those imagined futures rather than against the far more expensive real world [1, 2, 3]. Because the world is stochastic, a good world model must reproduce not merely the expected future but the whole distribution over futures, and the quality of that imagined distribution is what planning quality inherits (Figure 1). The past years have produced world models of remarkable breadth, from recurrent state-space models that support control purely from imagination [4, 3] to latent-dynamics models embedded in model-predictive control [5], autoregressive token-based dynamics [6], and generative video models that ren-

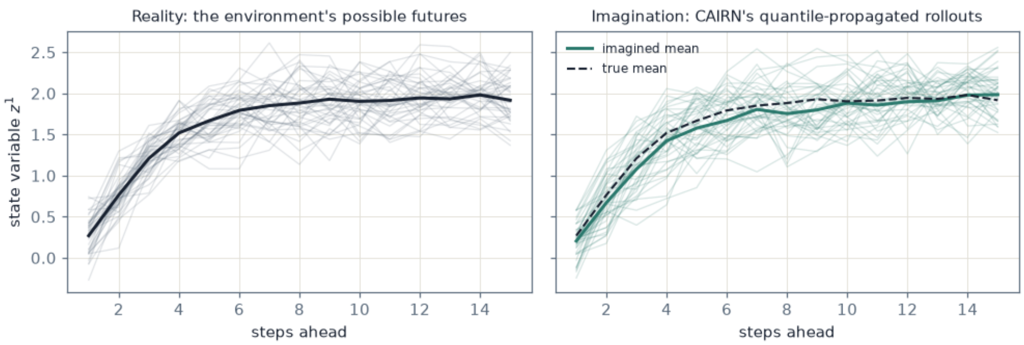

Figure 1: What a world model does. Left: forty futures that the real environment of Section 4 can produce from one state under one action plan. Right: forty futures CAIRN imagines for the same situation by propagating samples through its per-variable quantile heads. A useful world model must reproduce not only the mean of what will happen (dashed) but the spread of what could.

der entire interactive environments [7]. Despite their diversity, these models share one structural commitment: the transition function is a *monolithic map* $\hat{s}_{t+1} = f_\theta(s_t, a_t)$ in which every latent dimension depends on every other and the action conditions the whole map.

Because of this commitment, three well-documented failure modes follow. First, the models are *interventionally brittle*: when one mechanism of the environment changes—an object's mass, a joint's gain, the collateral rule of one financial protocol—the entire learned transition is off-support, so the model cannot localize the change, cannot adapt to it without retraining everything, and its rollouts degrade globally even though most of the world is unchanged. Second, they carry *no native validity semantics*: a monolithic model emits point predictions or heuristic ensemble variances [8], but nothing inside it can state, with controlled error, that a particular part of its dynamics is currently stale. Third, *actions are not treated as interventions*: physically and economically an action manipulates specific variables and its effects propagate through structure, yet conditioning a monolithic map on the action vector discards exactly the sparsity that compositional generalization to unseen action–state combinations requires [9, 10].

Recent causal world-model research addresses the third point by factoring the transition over a learned causal graph [11, 12], and interventional causal representation learning provides identifiability foundations for such factorizations [13]. What remains missing is any mechanism by which the model itself *notices*, during deployment and with statistical guarantees, that one of its mechanisms has become invalid—and then responds locally rather than globally. Post-hoc monitoring wrappers such as adaptive conformal inference [14] and online randomness testing [15] operate outside the model on scalar summaries; they can raise a global alarm but cannot say *which* mechanism broke, let alone repair it.

This paper closes that gap with CAIRN, a world model whose central design principle is that *the causal factorization and the validity monitoring must be built together*, because each is what makes the other useful. The factorization gives every latent variable its own small mechanism with a sparse parent set; the monitoring equips every mechanism with an internal betting martingale over its own predictive validity, in the tradition of game-theoretic statistics [16, 17, 18]. When a mechanism is valid, no betting strategy can grow wealth systematically; when it breaks, wealth grows exponentially, and by Ville's inequality [19] the wealth crossing a threshold $1/\delta$ is an anytime-valid, per-mechanism alarm at level δ that requires no tuning and no multiple-testing correction over continuous monitoring. Because the alarm names a specific mechanism, the response can be local: spawn a fresh copy of that mechanism, refit it on a short recent window, and leave the rest of the model untouched; persistent alarms after refit indicate a wrong parent set and trigger a local re-search of the graph.

Our contributions are as follows. First, we specify the CAIRN model class and training objective: a transition kernel factored over a learned adjacency and a learned action-target mask, distributional (quantile) mechanism heads, interventions as architectural graph surgery, and a joint loss combining quantile regression, sparsity, interventional consistency, a sparse-mechanism-shift invariance penalty, and a differentiable rollout calibration term (Section 3). Second, we develop the e-gate: a composite betting e-process over probability-integral-transform residuals with Online Newton Step bet sizing, a conformal recalibration step that restores finite-sample validity under learned quantile heads, and a *tolerance null* that keeps Ville's guarantee exact in the presence of irreducible approximation error; we prove time-uniform false-alarm control, bound detection delay, and give conditions under which alarms localize to the truly shifted mechanisms (Section 3.2). Third, we show how the gate's wealth acts inside the model: evidence-weighted mechanism mixtures, uncertainty inflation, and online structure repair, together with risk-sensitive planning through quantile rollouts (Section 3.4). Fourth, on a ground-truth synthetic benchmark with scriptable mechanism shifts we evaluate six pre-registered research questions against monolithic, ensemble, and change-detection baselines, reporting both the successes and the places where the causal machinery does not help (Section 4).

2. Related work

CAIRN stands at the intersection of three literatures that have so far developed largely independently of one another: world models for planning and control, causal representation and dynamics learning, and anytime-valid sequential inference. We review each in turn, and close by situating our contribution against the small set of works that combine any two of the three.

2.1. World models for planning and control

The idea that an agent should learn a model of its environment and plan inside it is among the oldest in reinforcement learning: Dyna [1] already interleaved learning, planning, and acting through a learned model, and PILCO [20] demonstrated that a probabilistic dynamics model—there a Gaussian process—could make policy search dramatically more data-efficient precisely because it propagated model uncertainty through imagined rollouts rather than trusting a point estimate. The modern deep-learning lineage begins with the recurrent world models of **(author?)** [2], in which a learned latent simulator was rich enough for a policy to be trained entirely inside it, and with PlaNet [4], which showed that latent-space planning from pixels was practical at scale. The Dreamer family refined this recipe into imagination-based actor–critic learning, culminating in a single configuration that masters a wide diversity of control domains [3]; TD-MPC2 couples learned latent dynamics with sampling-based model-predictive control and scales it across more than a hundred continuous-control tasks [5]; IRIS recasts dynamics learning as autoregressive prediction over discrete tokens [6]; and Genie learns generative, action-controllable video environments from unlabelled footage alone [7]. On the uncertainty side, PETS [8] established the probabilistic-ensemble recipe that remains the de facto standard: several bootstrapped dynamics networks whose disagreement serves as an epistemic signal during trajectory sampling.

For all their differences in representation—recurrent state, latent vector, token sequence, video frame—these systems share the monolithic transition map described in the introduction, and they inherit its consequences.

When one mechanism of the environment changes, there is no component of the model that corresponds to the changed mechanism, so there is nothing to detect, nothing to blame, and nothing to repair short of fine-tuning the whole network, with the catastrophic interference that Section 4 quantifies. Ensemble disagreement, the standard uncertainty tool, measures where the members were trained differently—not where the world has since changed—and our experiments confirm that it neither controls false alarms nor localizes genuine shifts. CAIRN differs from this entire line not in how well it predicts, but in what it is made of: mechanisms that can be individually monitored, individually distrusted, and individually replaced.

2.2. Causal representation and dynamics learning

That learned models should factor along causal lines is the central tenet of causal representation learning [9], whose core premise—the sparse-mechanism-shift hypothesis—holds that naturally occurring distribution shifts affect few of the underlying mechanisms at once. This premise has been used as a learning signal in several ways. Invariant causal prediction exploits the fact that only the true causal parents of a variable yield predictions that are stable across environments [21]; the meta-transfer objective of **(author?)** [22] identifies causal structure by preferring factorizations that adapt fastest to interventions; and interventional causal representation learning gives identifiability guarantees when data from genuine interventions are available [13]. On the dynamics side, causal dynamics learning recovers a sparse transition graph and uses it for task-independent state abstraction [11], Denoised MDPs separate controllable from uncontrollable factors of variation [12], and differentiable structure discovery descends from NOTEARS [23], whose continuous acyclicity penalty made graph learning amenable to gradient descent. Our setting sidesteps the least stable ingredient of that machinery: because a world model's edges run strictly forward in time, the graph is a dynamic Bayesian network and acyclicity holds by construction, so structure learning reduces to sparse masking with Gumbel-sigmoid relaxations [24, 25]. CausalWorld supplies a robotic benchmark whose ground-truth causal structure makes such claims testable [26]; our synthetic suite plays the same diagnostic role at a scale where every quantity of interest—graph, targets, descendants, shifted mechanisms—can be scored exactly.

What none of these works provide is any *deployment-time* account of mechanism validity. A causal factorization is learned, frozen, and then trusted; if a mechanism goes stale, the factored model fails in the same silent way a monolithic one does, merely along cleaner seams. The premise that shifts are sparse is used as a training-time regularizer, never as a run-time diagnostic. CAIRN's position is that the factorization earns its keep precisely at deployment: it is the addressing scheme that lets statistical evidence about "something is wrong" become actionable evidence about "*this* is wrong".

2.3. Anytime-valid inference and sequential monitoring

The statistical machinery we attach to each mechanism descends from the oldest rigorous treatment of gambling against a hypothesis: **(author?)** [19] introduced the inequality by which a nonnegative supermartingale exceeds $1/\delta$ with probability at most δ, turning betting wealth into a time-uniform certificate. This view has been developed into the modern theory of game-theoretic probability and safe anytime-valid inference [16, 17], in which e-processes replace p-values wherever data arrive sequentially and stopping rules are unknowable in advance. **(author?)** [18] showed that betting strategies sized by online convex optimization yield confidence sequences that are simultaneously valid and nearly optimal, with Online Newton Step [27] providing the logarithmic regret that our detection-delay bound leans on. Closest in spirit to our monitoring problem, conformal test martingales test exchangeability online [28, 15], and adaptive conformal inference tracks miscoverage of a deployed predictor under distribution shift [14]. Classical sequential change detection, embodied by the CUSUM procedure [29], remains the standard engineering answer and serves as our strongest tuned baseline. Finally, our residuals are probability integral transforms of distributional forecasts, whose role in calibration assessment is classical [30], and our mechanism heads are quantile regressions trained with the pinball loss [31].

Every one of these monitors, however, is *external*: it watches a scalar summary of a model from outside and can at best raise a global alarm. The distinctive move in CAIRN is architectural internalization. There is one e-process per mechanism, its input is that mechanism's own predictive validity, and its wealth is not a dashboard number but a model parameter: it reweights the mechanism library, inflates exactly that mechanism's rollout uncertainty, triggers its refit, and arbitrates repair of its parent set. To our knowledge, no prior world model carries anytime-valid evidence inside its own transition function.

2.4. Positioning

Combining two of the three literatures is not new: model-based reinforcement learning with ensembles combines world models with (heuristic) uncertainty [8]; causal dynamics learning combines world models with causal structure [11]; conformal monitoring combines deployed predictors with valid sequential inference [14, 15]. The three-way combination is, to our knowledge, new, and Section 4 is designed to show that it is not a decoration: the localization guarantee (Proposition 3) is available *only* to a factored model, the tolerance null (Proposition 1) is what makes betting guarantees survive learned mechanisms, and the ablation that removes structure—a single e-gate over a monolithic model—fails to detect localized shifts at all at matched tolerance.

3. The CAIRN world model

3.1. Model class

An encoder q_ϕ maps observations to a latent state partitioned into d variables $z_t = (z_t^1, \ldots, z_t^d)$. In the state-space instantiation studied experimentally the partition is given and the encoder is the identity; in pixel instantiations a frozen visual encoder such as DINOv2 [32] supplies the latent stream, and everything downstream is unchanged. Structure is carried by two learned binary matrices: an adjacency $A \in \{0,1\}^{d\times d}$, where $A_{ji} = 1$ means z_t^j is a parent of z_{t+1}^i, and an action-target mask $M \in \{0,1\}^{m\times d}$, where $M_{ki} = 1$ means action coordinate k acts on variable i. Both are parameterized by Gumbel-sigmoid relaxations with straight-through gradients [24, 25]. Because edges run from t to $t{+}1$, the graph is a dynamic Bayesian network and no acyclicity penalty is needed.

The transition kernel factorizes over mechanisms,

$$p_\theta\big(z_{t+1} \mid z_t, a_t\big) \;=\; \prod_{i=1}^{d} p_{\theta_i}\big(z_{t+1}^i \mid z_t \odot A_{\cdot i},\; a_t \odot M_{\cdot i}\big), \tag{1}$$

where each mechanism f_i is a small multilayer perceptron whose input is masked elementwise by its parent and target columns, so structure gradients flow through the masks. Every mechanism carries a *distributional head*: it outputs a vector of K quantiles $\{\hat{z}_{t+1}^i(\tau_k)\}_{k=1}^K$ at fixed levels $\tau_1 < \cdots < \tau_K$,

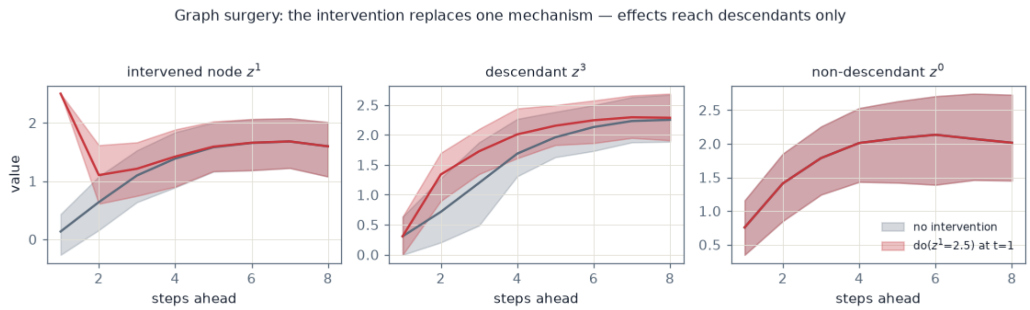

Figure 2: Graph surgery in imagination on the ground-truth benchmark of Section 4. The query do(z^1=2.5) pins the intervened variable at the surgery step (left), shifts the distribution of its descendant (middle), and leaves the non-descendant's imagined future exactly unchanged—the intervention band coincides with the no-intervention band (right). A monolithic model cannot make this promise.

made non-crossing by accumulating softplus-positive increments, and trained with the pinball loss [31]

$$\mathcal{L}_i^{\text{pin}} = \frac{1}{K}\sum_{k=1}^{K}\rho_{\tau_k}\big(z_{t+1}^i - \hat{z}_{t+1}^i(\tau_k)\big), \qquad \rho_\tau(u) = \max\{\tau u,\, (\tau-1)u\}. \quad (2)$$

Rollouts propagate samples drawn by inverting the piecewise-linear quantile function, so imagination is a distribution over trajectories and predictive intervals at every horizon come for free.

Interventions are first-class inputs. An intervention do($z^i \leftarrow v$)—an agent action reaching variable i through M, or an exogenous labeled shock—*replaces* mechanism i's output by the substituted value rather than conditioning it. This is the graph-surgery semantics of the do-operator [10] implemented architecturally: parents severed, value substituted. Counterfactual rollouts are therefore a forward pass. A direct consequence, used repeatedly below, is that variables that are not descendants of the intervened node are provably untouched by the intervention in the model's imagination (Figure 2).

3.2. The e-gate

The e-gate converts each observed transition into evidence about each mechanism's validity. Let F_t^i denote the predictive distribution that mechanism i stated for z_{t+1}^i (the piecewise-linear distribution through its quantiles), and let

$$u_t^i \;=\; F_t^i\big(z_{t+1}^i\big) \in (0,1) \tag{3}$$

be the (randomized-tail) probability integral transform of the realized value. If the stated distribution is the true conditional law, then u_t^i is uniform on $(0,1)$ conditionally on the past [30]. Mechanism i bets against this uniformity with a wealth process

$$W_t^i \;=\; \prod_{s\le t}\Big(1+\lambda_s^i\,\big(g(u_s^i)-\varepsilon\,\mathrm{sign}\,\lambda_s^i\big)\Big), \qquad W_0^i=1, \tag{4}$$

where g is a bounded payoff with $\mathbb{E}[g(U)]=0$ under uniformity, $\lambda_s^i \in [-\frac{1}{2},\frac{1}{2}]$ is a predictable bet, and $\varepsilon \ge 0$ is a tolerance whose role is explained below. We run two elementary payoffs per gate, a location bet $g_1(u)=2u-1$ and a dispersion bet $g_2(u)=6(u-\frac{1}{2})^2-\frac{1}{2}$, both bounded in $[-1,1]$, and combine the two wealth processes by averaging, which preserves the e-process property. Bets are sized by Online Newton Step on the log-wealth,

$$\lambda_{t+1} \;=\; \Pi_{[-\frac{1}{2},\frac{1}{2}]}\Big(\lambda_t+\tfrac{2}{2-\ln 3}\,\tfrac{\nabla_t}{\sum_{s\le t}\nabla_s^2}\Big), \qquad \nabla_t=\frac{g(u_t)-\varepsilon\,\mathrm{sign}\,\lambda_t}{1+\lambda_t(g(u_t)-\varepsilon\,\mathrm{sign}\,\lambda_t)}, \tag{5}$$

following **(author?)** [18]. The gate *alarms* the first time $W_t^i \ge 1/\delta$.

Figure 3 traces one transition through this machinery: the observation is partitioned into variables, each mechanism sees only the inputs its learned masks admit, predicts a full distribution, watches where reality lands inside that distribution, and places its bets. When the mechanism is healthy the PIT falls unpredictably and the bets lose money on average; when the mechanism is broken the PITs pile up in a region of the unit interval and the wealth compounds exponentially.

Two practical refinements make the guarantee hold for *learned* mechanisms rather than only for ideal ones. First, before deployment the raw PIT values are passed through a conformal recalibration map fitted on held-out data—the randomized empirical distribution function of calibration PITs—which makes the deployed PITs exactly uniform under exchangeability with the calibration set regardless of quantile-head bias; the same sliding-holdout recalibration runs periodically during deployment for gate-quiet mechanisms. Second, the tolerance $\varepsilon > 0$ in (4) converts the null from exact uniformity to the composite null "approximately valid", which is the operationally correct null when mechanisms carry irreducible approximation error: a learned

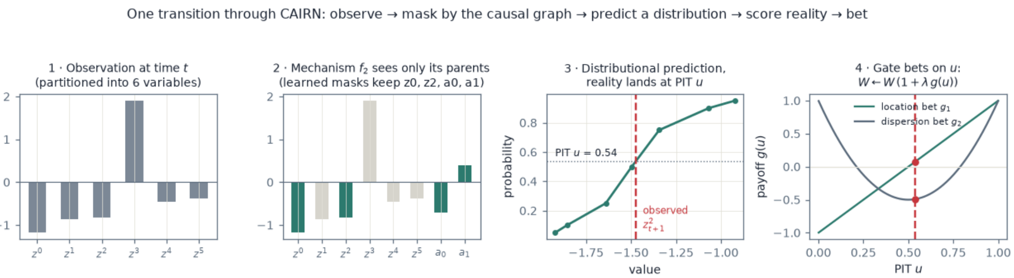

Figure 3: One transition through CAIRN. (1) The observation is partitioned into variables. (2) Mechanism f_2 receives only the inputs its learned causal masks admit—here two parent variables and both action coordinates; all other inputs are zeroed. (3) The mechanism predicts a full distribution through seven quantiles, and the realized next value lands at PIT $u = 0.54$, near the middle, as it should when the mechanism is valid. (4) The gate evaluates its location and dispersion payoffs at u and updates its wealth by (4).

model that is wrong by a hair should not be flagged as broken, while a mechanism whose physics changed produces payoffs far outside the tolerance and is caught within tens of steps.

Proposition 1 (Anytime-valid false-alarm control)**.** *Fix $\delta \in (0,1)$ and $\varepsilon \geq 0$, and let $(u_t)_{t\geq 1}$ be any sequence adapted to a filtration $(\mathcal{F}_t)$ such that $\big|\mathbb{E}[g(u_t) \mid \mathcal{F}_{t-1}]\big| \leq \varepsilon$ for all t. Then the wealth (4) with any predictable bets $\lambda_t \in [-\frac{1}{2}, \frac{1}{2}]$ satisfies $\mathbb{P}\big(\exists t : W_t \geq 1/\delta\big) \leq \delta$.*

Proof. Write $Y_t = \lambda_t\big(g(u_t) - \varepsilon \operatorname{sign} \lambda_t\big)$. Since λ_t is $\mathcal{F}_{t-1}$-measurable,

$$\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = \lambda_t \, \mathbb{E}[g(u_t) \mid \mathcal{F}_{t-1}] - \varepsilon|\lambda_t| \;\leq\; |\lambda_t|\Big(\big|\mathbb{E}[g(u_t) \mid \mathcal{F}_{t-1}]\big| - \varepsilon\Big) \;\leq\; 0.$$

Moreover $Y_t \geq -\frac{1}{2}(1+\varepsilon) > -1$, so $1 + Y_t > 0$ and $\mathbb{E}[1 + Y_t \mid \mathcal{F}_{t-1}] \leq 1$; hence $W_t = \prod_{s\leq t}(1 + Y_s)$ is a nonnegative supermartingale with $W_0 = 1$. Ville's inequality [19] gives $\mathbb{P}(\sup_t W_t \geq 1/\delta) \leq \delta$. The average of the two payoffs' wealth processes is again a nonnegative supermartingale with initial value one, so the composite gate inherits the same bound. □

Proposition 2 (Detection delay)**.** *Suppose that after an unannounced change the payoffs are independent and identically distributed with $\mu = \mathbb{E}[g(u_t)]$ satisfying $|\mu| > \varepsilon$. Let $r = \dfrac{(|\mu| - \varepsilon)^2}{4(1+\varepsilon)^2}$. Then the gate with the fixed oracle bet $\lambda^* = \operatorname{sign}(\mu)\, \dfrac{|\mu| - \varepsilon}{2(1+\varepsilon)^2}$ satisfies $\liminf_{t\to\infty} \frac{1}{t} \log W_t \geq r$ almost surely, so its* *alarm time is at most* $(1 + o(1)) \log(1/\delta)/r$*; and the ONS gate* (5) *attains the same asymptotic rate, since its log-wealth is within* $O(\log t)$ *of the best constant bet in hindsight.*

Proof. Set $s = \operatorname{sign}(\mu)$ and $Y_t = \lambda^*(g(u_t) - \varepsilon s)$. Then $|Y_t| \leq |\lambda^*|(1+\varepsilon) \leq \frac{1}{2}$, and for $|y| \leq \frac{1}{2}$ one has $\log(1 + y) \geq y - y^2$. Hence

$$\mathbb{E} \log(1 + Y_t) \;\geq\; \mathbb{E}Y_t - \mathbb{E}Y_t^2 \;\geq\; |\lambda^*|(|\mu| - \varepsilon) - |\lambda^*|^2(1+\varepsilon)^2 \;=\; r,$$

where the last equality follows by substituting λ^*. By the strong law of large numbers applied to the i.i.d. bounded variables $\log(1 + Y_t)$, $\frac{1}{t} \log W_t(\lambda^*) \to \mathbb{E} \log(1+Y_1) \geq r$ almost surely, and $W_t \geq 1/\delta$ once $t \geq (1+o(1)) \log(1/\delta)/r$. The losses $-\log(1 + \lambda(g_t - \varepsilon \operatorname{sign}\lambda))$ are exp-concave in λ on the compact bet space, so Online Newton Step guarantees $\log W_t(\text{ONS}) \geq \log W_t(\lambda^*) - O(\log t)$ [27], which does not affect the leading term. □

Proposition 3 (Localization)**.** *Consider a shift confined to a mechanism subset* S*, and suppose that for every* $i \notin S$ *the stated predictive distribution of mechanism* i *remains within the betting tolerance, in the sense that* $\left|\mathbb{E}[g_k(u_t^i) \mid \mathcal{F}_{t-1}]\right| \leq \varepsilon$ *for both payoffs and all* t*. Then every gate outside* S *alarms with probability at most* δ *over the entire deployment, while every gate* $i \in S$ *whose post-change payoff mean exceeds the tolerance by* $\Delta_i = |\mu_i| - \varepsilon > 0$ *alarms within* $(1 + o(1))\, 4(1+\varepsilon)^2 \log(1/\delta)/\Delta_i^2$ *steps. Family-wise control over the* $d - |S|$ *healthy gates holds at level* $(d - |S|)\delta$ *by the union bound, or at level* δ *by merging the per-gate e-values by averaging.*

Proof. The monitoring inputs of gate i are functions of the *realized* transition (z_t, a_t, z_{t+1}^i) only: predictions are conditioned on the observed state, never on imagined values. A shift of the mechanisms in S changes the law of z_{t+1}^j for $j \in S$ and, through the state, the distribution of *inputs* visited by other mechanisms; but the conditional law of z_{t+1}^i given (z_t, a_t) for $i \notin S$ is unchanged, and by hypothesis the stated distribution remains within tolerance on the visited inputs. Proposition 1 then applies to each healthy gate separately, giving the per-gate bound; the union bound and the fact that an average of e-values is an e-value give the two family-wise statements. The delay claim is Proposition 2 applied to each $i \in S$. □

Remark 1. The hypothesis of Proposition 3 is exactly where honest caveats live: a healthy mechanism can leave its tolerance when the shifted state distribution drives it into input regions where its approximation error is larger than ε. Empirically (Section 4.2) this collateral effect is rare and mild, and the periodic recalibration of gate-quiet mechanisms absorbs slow drifts of this kind, at the documented price that sub-tolerance drifts are absorbed rather than flagged.

3.3. Training objective

CAIRN is trained on trajectory data spanning multiple regimes $e \in \mathcal{E}$, each regime differing from the nominal one by a sparse mechanism shift, together with a fraction of steps carrying labeled do-interventions. The loss combines five terms,

$$\mathcal{L} = \sum_{i=1}^{d} \mathcal{L}_i^{\text{pin}} + \gamma_1\big(\|A\|_1 + \|M\|_1\big) + \gamma_2\, \mathcal{L}^{\text{int}} + \gamma_3\, \mathcal{L}^{\text{inv}} + \gamma_4\, \mathcal{L}^{\text{cal}}, \tag{6}$$

whose parts we describe in turn. The pinball term $\mathcal{L}^{\text{pin}}$ is the teacher-forced one-step loss (2) plus a multi-step latent-overshooting variant in which the predicted median is propagated through short segments and penalized at every step, fighting compounding error. The sparsity term applies an ℓ_1 penalty to the edge and target probabilities. The interventional-consistency term $\mathcal{L}^{\text{int}}$ applies the same multi-step pinball loss on segments containing labeled interventions, computed through the do-surgery forward pass, so that both A and M receive interventional rather than merely observational signal—the ingredient that identifiability results require [13]. The invariance term

$$\mathcal{L}^{\text{inv}} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \sum_{i=1}^{d} \sqrt{\,\big(\bar{r}_i^{\,e} - \bar{r}_i\big)^2 + \big(\sigma_i^{e} - \sigma_i\big)^2 + \epsilon} \tag{7}$$

penalizes, with a concave square-root relaxation of a count, the number of mechanisms whose standardized residual distributions differ across regimes: a wrong factorization spreads any regime shift across many pseudo-mechanisms and pays more than the true factorization, which concentrates it [9]. The calibration term $\mathcal{L}^{\text{cal}}$ is a smoothed-indicator coverage penalty on multi-step intervals,

$$\mathcal{L}^{\text{cal}} = \sum_{(k_1,k_2)} \Big(\mathbb{E}\big[\sigma_s(z - \hat{z}(\tau_{k_1}))\, \sigma_s(\hat{z}(\tau_{k_2}) - z)\big] - (\tau_{k_2} - \tau_{k_1})\Big)^2, \tag{8}$$

with σ_s a temperature-s sigmoid, so calibration is optimized during training rather than patched afterwards. Optimization is two-timescale: mechanism parameters move at a fast learning rate, structure logits at a slower effective timescale with a delayed start and Gumbel temperature annealing.

Before monitoring begins in a deployment regime, mechanisms are refit to that regime with the learned structure frozen (regime-entry adaptation), and PIT recalibration is fitted on held-out deployment-regime data; the gates then certify *continued* validity, which is their job. This train–monitor separation is enforced throughout: wealth carries guarantees only on data never used for fitting.

3.4. Deployment: adaptation, inflation, repair, planning

Wealth enters the model in three places, which is what makes the e-gate an algorithmic component rather than a monitor. First, each variable keeps a small library $\{f_i^{(1)}, \dots, f_i^{(K_i)}\}$ of mechanism copies; rollouts combine the members' quantile functions with evidence weights

$$\pi_i^{(k)} \;\propto\; \exp\bigl(-\beta\, \log W_t^{i,(k)}\bigr), \tag{9}$$

so members accumulating evidence of invalidity are smoothly down-weighted. When a gate alarms, a fresh copy is spawned and fitted on the earlier three quarters of a short recent window (the held-out tail is reserved for recalibrating the new gate), which is a few-shot problem by construction because each mechanism is small and its parent set sparse; the rest of the graph is untouched. Second, during h-step imagination each mechanism's predictive spread is inflated around its median by a factor

$$c_i \;=\; \min\Bigl\{c_{\max},\; 1 + \alpha\, \frac{\max(0, \log W_t^i)}{\log(1/\delta)}\Bigr\}, \tag{10}$$

so an invalid-looking mechanism contributes honest extra uncertainty to exactly the variable it generates, propagating structurally to descendants (Figure 7). Third, persistent alarms at a node *after* refit indicate a wrong parent set and trigger a local re-search over $A_{\cdot i}$: candidate single-edge flips are scored by held-out pinball loss on the recent window and adopted when they beat the current set by a margin—online causal-structure repair driven by anytime-valid evidence. Planning uses any sampling-based planner on quantile-propagated returns; we use the cross-entropy method scored by conditional value-at-risk, so candidate action sequences whose effects flow through low-wealth (healthy) mechanisms are preferred and the planner routes around the broken part of the world model without any dedicated machinery. Algorithm 1 summarizes the loop, and Figure 4 the architecture.

Algorithm 1 CAIRN deployment loop

offline: train $(q_\phi, \{f_i\}, A, M)$ with (6) on multi-regime data; regime-entry refit; fit PIT recalibrators on held-out data
for $t = 1, 2, \ldots$ **do**
  $z_t \leftarrow q_\phi(o_t)$
  $a_t \leftarrow$ CEM over CVaR of quantile rollouts with weights (9) and inflation (10)
  execute a_t; observe z_{t+1}
  for $i = 1, \ldots, d$ **do**
    $u \leftarrow$ recalibrated PIT of z^i_{t+1} under mechanism i
    update W^i by (4) and λ^i by (5)
    if $W^i \geq 1/\delta$ **then** ▷ anytime-valid local alarm
      spawn $f_i^{(K_i+1)}$; few-shot refit on recent window; fresh gate
      if repeated alarms at i **then** local re-search over $A_{\cdot i}$
      end if
    end if
  end for
  every T_{recal} steps: refit recalibrators and restart gates of gate-quiet mechanisms ▷ sliding-holdout maintenance
end for

4. Experimental evaluation

4.1. Benchmark, baselines, and protocol

The claims of this paper are claims about *correctness*: that alarms respect a false-alarm guarantee, that they name the truly shifted mechanisms, that non-descendants of an intervention are untouched, and that adaptation repairs what broke and nothing else. Claims of this kind can only be scored against an environment whose causal structure, shift locations, and interventional ground truth are known exactly, which is why our primary testbed is synthetic by design rather than by convenience—the same diagnostic role that CausalWorld plays for robotic setups [26], at a scale where every quantity of interest is computable in closed form. The environment is a nonlinear dynamic Bayesian network: $d = 8$ variables, $m = 3$ action coordinates, mechanisms $z^i_{t+1} = \rho_i z^i_t + \tanh(w_i^\top z_t^{\mathrm{pa}(i)} + c_i^\top a_t^{\mathrm{tg}(i)} + b_i) + \sigma\xi$ with edge weights bounded away from zero, known adjacency and action targets, scriptable per-regime mechanism shifts, and do-interventions on individual

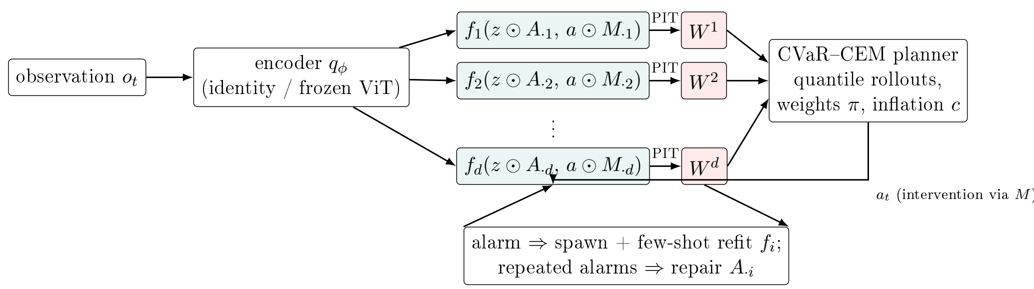

Figure 4: CAIRN architecture. Observations are encoded into a factored latent state; each variable's mechanism sees only its masked parents and action targets and outputs quantiles. Each mechanism's PIT residuals feed its own betting wealth W^i, which weights the mechanism library, inflates rollout uncertainty, and—upon anytime-valid alarms—triggers localized refits and structure repair. The planner acts through quantile rollouts and thus routes around unhealthy mechanisms.

variables (Figure 5). Training data span four regimes (nominal plus three sparse shifts) with labeled interventions restricted to variables $\{0, 1, 2, 3\}$, so that do-queries on the remaining variables are compositionally held out: at evaluation time the model is asked what happens under interventions of a kind it has seen applied only to *other* variables. Before deployment-time monitoring begins, every model undergoes the regime-entry refit of Section 3.3, and the e-gates receive their conformal calibration on held-out deployment-regime episodes, so that all guarantees are evaluated on data disjoint from everything used for fitting.

The comparison set is chosen so that each claimed capability has a baseline designed to take it away. Against prediction and planning we compare a monolithic quantile world model of matched training budget and architecture family—the canonical DreamerV3/TD-MPC2-style commitment, instantiated in state space so that the comparison isolates factorization rather than encoder quality—and a five-member probabilistic ensemble with trajectory sampling in the PETS tradition [8]. Against detection we compare the classical CUSUM procedure [29] run per variable on standardized residuals, granted *oracle* tuning: its thresholds are set to the maximum statistic observed on a held-out null stream, the most favourable calibration a practitioner could perform, and one that CAIRN, which needs no tuning at all, must beat anyway. Ensemble disagreement with the same oracle thresholding tests the folk wisdom that epistemic uncertainty suffices for shift detection. A global e-gate over the monolithic model—identical statistical machinery, no

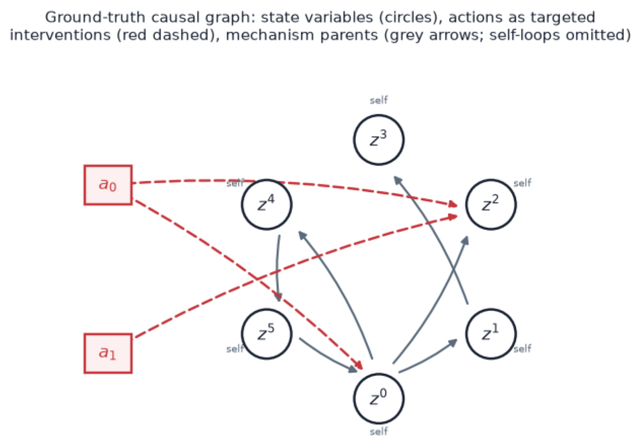

Figure 5: A six-variable instance of the ground-truth benchmark family used for illustration throughout: state variables (circles) with sparse parent edges (grey), and action coordinates entering as targeted interventions (dashed). The evaluation suite uses $d = 8$ variables with in-degree two.

factorization—isolates the role of structure, and the CAIRN ablations (oracle graph as an upper bound; no invariance term) map each component to its contribution. All experiments run on a CPU workstation; the complete suite trains in under an hour, and every number below regenerates from committed scripts.

4.2. Detection and localization (RQ2)

Table 1 contains the headline result, aggregated over three independent benchmark seeds (fresh graphs, mechanisms, and data per seed); Figure 6 shows a representative episode. Across ten unannounced shift scenarios per seed (eight single-mechanism sign flips and two two-mechanism shifts), the e-gates detect the shifted mechanisms with a median delay of 17.2 ± 0.6 steps and localize them with an F_1 of 0.90 ± 0.07; over three 6,000-step stationary null streams per seed they raise *zero* false alarms on every seed, consistent with (and conservatively below) the guarantee of Proposition 1, which the exact-null verification confirms independently at an alarm fraction of 0.0000. The comparison detectors illustrate both classical failure modes. Oracle-tuned CUSUM is sometimes instantaneous but wildly inconsistent: its thresholds, tuned to zero false alarms on one null stream, produce a 0.50 ± 0.09

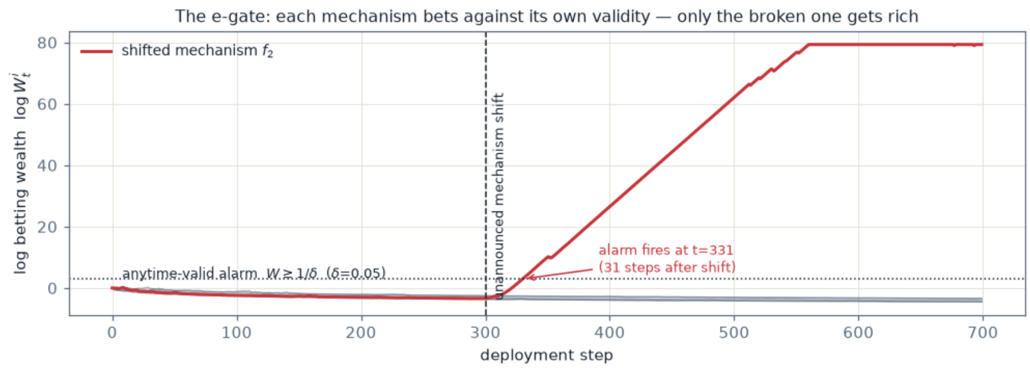

Figure 6: The e-gate catching a broken mechanism. At step 300 one mechanism's dynamics are silently sign-flipped. Healthy mechanisms' log wealth stays near zero (grey); the broken mechanism's wealth grows exponentially and crosses the anytime-valid threshold 31 steps later. The alarm names exactly the variable whose physics changed.

per-gate false-alarm fraction on fresh null streams, and its localization F_1 of 0.69 ± 0.12 mixes perfect scenarios with complete misses. Ensemble disagreement names the wrong variables ($F_1 = 0.29 \pm 0.06$) and false-alarms on half the gates. The global e-gate on the monolithic model—the ablation that removes structure—is the starkest contrast: running at the same tolerance, it never detects any of the ten shifts within the 900-step window, because the evidence of one broken mechanism among eight is diluted below the tolerance when averaged across the whole model. Localization is therefore not a bonus on top of detection; at realistic tolerances, per-mechanism factorization is what makes detection possible at all.

4.3. Interventional generalization (RQ1)

Table 2 evaluates off-support do-queries ($\mathrm{do}(z^i = \pm 4)$, far outside the reachable range) on held-out and trained do-targets. The structural signature is unambiguous: with the oracle graph, non-descendant error under intervention *equals* the no-intervention reference to three decimals, as Proposition-level reasoning predicts and Figure 2 visualizes; the learned graph leaks almost nothing (e.g. 0.580 against a reference of 0.552 on trained targets); the monolithic model leaks severely (0.708 against 0.456, a 55% violation), and the ensemble similarly. On raw rollout RMSE, by contrast, all model classes are statistically tied at long horizons: in this stochastic, multi-attractor environment, long-horizon error is dominated by attractor-branching uncertainty

Table 1: Detection and localization (RQ2): mean ± standard deviation over 3 independent benchmark seeds, each aggregating ten unannounced mechanism-shift scenarios and three 6,000-step stationary null streams at level $\delta = 0.05$. The e-gates run with no tuning; CUSUM and ensemble thresholds receive oracle tuning on a separate null stream. The exact-null verification (gates fed probability integral transforms from the true environment conditional) produced an alarm fraction of 0.0000, consistent with Proposition 1. *per stream rather than per gate.

Detector	Null alarm fraction	Median delay	Localization F_1
CAIRN e-gates	0.000 ± 0.000	17.167 ± 0.624	0.904 ± 0.066
e-gates, oracle graph	0.000 ± 0.000	20.333 ± 3.325	0.911 ± 0.042
CUSUM	0.500 ± 0.090	1.667 ± 0.471	0.693 ± 0.123
Ensemble disagreement	0.514 ± 0.255	49.333 ± 39.721	0.286 ± 0.061
Global e-gate (monolithic)	0.000 ± 0.000*	900.000 ± 0.000	undefined

that afflicts every model equally. We report this honestly rather than claiming a rollout-accuracy advantage; the causal factorization buys *invariance* and everything built on it (localization, adaptation, honest uncertainty), not a smaller mean-squared error on this benchmark.

4.4. Localized adaptation (RQ3)

If detection is to matter, it must change what adaptation costs. The promise of a factored model is that repairing a one-mechanism shift should be a one-mechanism problem—a few dozen samples for a small network with a sparse parent set—whereas a monolithic model must relearn a joint map in which the shift is entangled with everything it got right. Table 3 tests exactly this: it compares evidence-triggered localized refits against full fine-tuning of either CAIRN or the monolithic model at a matched gradient budget on the same post-shift stream, scoring both the shifted variables and, crucially, the healthy ones. The localized refit improves the shifted mechanisms steadily while the error of the seven healthy mechanisms is untouched—overall error *decreases* throughout adaptation. Both fine-tuning baselines suffer catastrophic interference: mid-adaptation, the monolithic model's overall error is an order of magnitude above its pre-shift level, and after four hundred samples it remains several times worse than before the shift. No method reaches the strict pre-shift error level within the window—the refit window of 96 samples bounds the achievable fit—which we report as a limitation of few-shot refitting rather than hiding it in a favorable metric.

Table 2: Interventional generalization (RQ1). Rollout RMSE of the model-mean trajectory against the true conditional mean under off-support do-interventions ($\mathrm{do}(z^i = \pm 4)$), for do-targets never intervened during training (held-out) and for trained do-targets, together with the structural split at horizon 3: error on descendants versus non-descendants of the intervened variable, and the no-intervention reference error of the same non-descendant set. A structurally correct model must leave non-descendants unchanged, so its non-descendant column must match the reference column.

Targets	Model	h=1	h=3	h=5	desc.	non-desc.	ref.
held-out	CAIRN (learned graph)	0.349	1.025	1.748	0.657	0.758	0.739
held-out	CAIRN w/o $\mathcal{L}^{\text{inv}}$	0.376	1.065	1.791	0.649	0.809	0.815
held-out	CAIRN (oracle graph)	0.412	1.017	1.640	0.616	0.744	0.744
held-out	Monolithic WM	0.328	0.916	1.524	0.607	0.757	0.726
held-out	Probabilistic ensemble	0.346	1.017	1.715	0.759	0.790	0.745
trained	CAIRN (learned graph)	0.313	0.990	1.405	0.885	0.580	0.552
trained	CAIRN w/o $\mathcal{L}^{\text{inv}}$	0.342	1.003	1.445	0.912	0.595	0.605
trained	CAIRN (oracle graph)	0.379	0.998	1.403	0.704	0.671	0.671
trained	Monolithic WM	0.276	1.105	1.594	1.043	0.708	0.456
trained	Probabilistic ensemble	0.288	1.061	1.521	1.089	0.616	0.476

4.5. Calibration (RQ4)

A world model that cannot be sure should say so, and it should say so about the right variables. Calibration is therefore not a cosmetic property here but the interface between monitoring and planning: the planner consumes predictive intervals, and whether those intervals are honest—before a shift, during the window in which a shift has occurred but has not yet been repaired, and after adaptation—determines whether risk-sensitive planning means anything at all. Table 4 reports empirical coverage of nominal 90% rollout intervals across exactly those three phases. In distribution, CAIRN's intervals are near-nominal across horizons where the ensemble under-covers. Under an unannounced shift and before any adaptation, the stale intervals lose most of their coverage on the shifted variable (down to 0.42 at horizon one against a nominal 0.90); evidence-driven inflation (10) restores honesty precisely there, at the cost of width, and after localized adaptation coverage recovers substantially on the refit mechanism (Figure 7). This is the operational meaning of "knowing where the imagined future is untrustworthy".

4.6. Structure recovery and regime diversity (RQ6)

Everything above presupposes that the structure CAIRN learns is close to the truth, and the theory of interventional identifiability predicts that the

Table 3: Localized adaptation versus fine-tuning (RQ3): one-step median RMSE after an unannounced shift of mechanism set S, as a function of the number of post-shift samples n, evaluated on the shifted variables ("shifted") and on all variables ("all"). All methods receive the same stream and the same gradient budget. Full fine-tuning suffers catastrophic interference on the seven healthy mechanisms; the evidence-triggered localized refit leaves them untouched.

Shift	Method	Nodes	n=0	n=25	n=50	n=100	n=200	n=400
S={1}	CAIRN localized refit	shifted	0.819	0.819	0.541	0.541	0.358	0.360
		all	0.347	0.347	0.270	0.270	0.229	0.229
S={1}	CAIRN full fine-tune	shifted	0.819	0.489	0.459	0.435	0.327	0.256
		all	0.347	1.899	2.497	2.824	0.770	0.681
S={1}	Monolithic fine-tune	shifted	0.743	0.972	5.026	4.656	0.693	0.694
		all	0.332	1.456	3.310	3.176	0.967	1.163
$S=\{3,5\}$	CAIRN localized refit	shifted	1.495	1.572	1.571	1.304	1.312	0.449
		all	0.771	0.809	0.808	0.793	0.802	0.331
$S=\{3,5\}$	CAIRN full fine-tune	shifted	1.495	0.576	2.584	2.152	1.318	1.460
		all	0.771	0.828	1.552	1.915	1.622	1.039
$S=\{3,5\}$	Monolithic fine-tune	shifted	1.460	1.495	1.179	1.413	2.108	2.330
		all	0.802	1.343	2.507	2.936	2.396	2.260

decisive resource is not more data but more *diverse* data: regimes that shift different mechanisms and interventions that strike different targets. Table 5 tests this prediction directly, showing recovery of A and M as the number of training regimes grows while the total number of episodes is held fixed, so that diversity varies without sample size varying. Action-target recovery improves markedly with regime diversity (F_1 from 0.57 with a single regime to 0.89 with four or more), in line with the identifiability role of interventional variation [13]; adjacency recovery is high throughout, with residual errors concentrated on edges whose influence approaches the noise floor. Downstream planning (RQ5) shows parity with the monolithic baseline in the nominal regime and a directional advantage for the adapted model after shifts, but with only six evaluation episodes per condition the intervals are wide, and we treat these planning numbers as preliminary.

5. Conclusion

This paper began with a robot whose failing wrist bearing left no trace in its own world model, and asked what it would take for that sentence to become impossible. The answer we have defended is architectural: give every

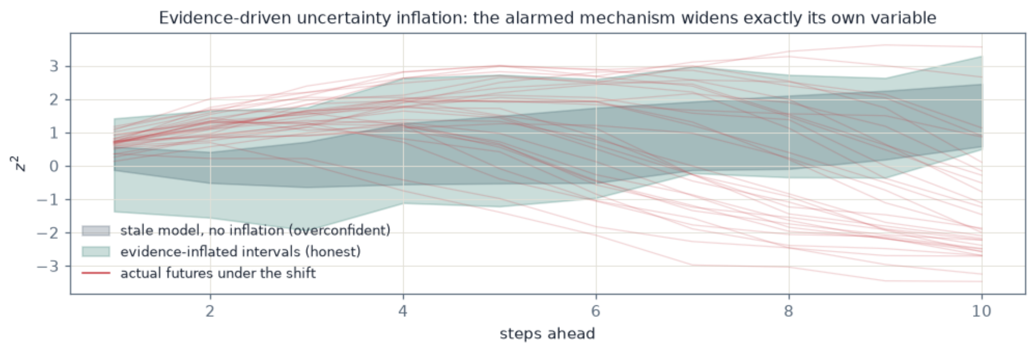

Figure 7: Evidence-driven uncertainty inflation on the shifted variable. The stale mechanism's old intervals (grey) miss the actual post-shift futures (thin lines); inflating the spread of exactly the alarmed mechanism (shaded, capped by (10)) restores honest coverage without touching healthy variables.

Table 4: Calibration (RQ4): empirical coverage of nominal 90% rollout intervals by horizon. In-distribution rows average over all variables; shift rows are scored on the shifted mechanism's variable when available (all-node averages otherwise), where interval honesty is actually at stake.

Condition	Scope	h=1	h=5	h=10	h=15
in-distribution	CAIRN (learned graph)	0.964	0.947	0.927	0.941
in-distribution	Probabilistic ensemble	0.880	0.841	0.859	0.879
shift, stale intervals	shifted node	0.417	0.490	0.792	0.719
shift, evidence-inflated	shifted node	0.771	0.771	0.938	0.969
shift, ensemble	all nodes	0.806	0.759	0.823	0.785
shift, after adaptation	shifted node	0.635	0.771	0.740	0.802

latent variable its own mechanism over a learned sparse parent set, treat actions and shocks as surgery on those mechanisms, and make each mechanism carry the statistical means to convict itself. We introduced CAIRN, a world model that couples a causally factored transition kernel with per-mechanism anytime-valid betting martingales, and showed that the joint construction produces a qualitatively new capability: a model that degrades *locally and detectably* under mechanism shifts instead of globally and silently. Neither half suffices alone—the ablations show that the statistical machinery without structure cannot even detect localized shifts at realistic tolerances, while structure without monitoring fails as silently as any monolith. The e-gate's guarantees are simple, tuning-free, and—with conformal PIT recalibration

Table 5: Structure recovery versus regime diversity (RQ6): structural Hamming distance and edge F_1 of the learned adjacency A and action-target mask M against ground truth, as the number of training regimes grows.

Regimes	SHD(A)	$F_1(A)$	SHD(M)	$F_1(M)$
1	4	0.889	3	0.571
2	4	0.889	3	0.571
4	8	0.800	1	0.889
6	6	0.842	1	0.889

and a tolerance null—hold for learned mechanisms rather than only in theory; on the ground-truth benchmark the gates achieved zero false alarms while localizing real shifts within tens of steps, where oracle-tuned classical detectors and ensembles fail in one direction or the other. Evidence-triggered adaptation repaired exactly the broken mechanisms without interference, and evidence-proportional inflation kept imagined futures honest while stale.

The road from the diagnostic benchmark to physical instantiations is short, and the ingredients are already in place (Figure 8). The actuator-damage scenario that our benchmark abstracts—one mechanism silently losing its gain—is literally available in standard robotic simulators: the top panel of Figure 8 shows an episode of the DeepMind Control walker [33] in which the right-knee actuator loses 95% of its torque mid-episode, visible in the collapse of exactly one proprioceptive trace while its left counterpart keeps full range; each such trace is precisely what one CAIRN mechanism would own, and the asymmetry between them is precisely what the e-gates detect and localize. For pixel-based deployment the bottom panel demonstrates the frozen-encoder pathway on real street video from the KITTI dataset [34]: pushing every frame through a frozen DINOv2 encoder [32] and projecting to a few principal components yields a smooth, low-dimensional latent stream extracted from raw video with no training at all, and it is over such streams—exactly as over the proprioceptive ones—that the factored mechanisms and their gates operate unchanged. Real-robot trajectory corpora such as DROID [35] then provide the natural testbed for offline evaluation of PIT validity and calibration on real data.

Several limitations bound these claims, and we state them together with the conclusions they qualify. The primary instantiation is in state space; pixel-based instantiations are architecturally prepared through a frozen visual encoder, but not yet evaluated. The tolerance null makes approximation error unflaggable below ε by design, and the sliding-holdout maintenance that keeps healthy gates quiet can absorb drifts below the tolerance; both are the operationally correct compromise, and both are stated on the tin. On raw long-horizon rollout accuracy the causal factorization brought no advantage in our multi-attractor benchmark, and the planning gains after adaptation, while directionally consistent, need many more evaluation episodes. Finally, the detection and localization results are aggregated over three independent benchmark seeds, while the remaining research questions report a single seed; full multi-seed intervals for every question, the robotic instantiations on CausalWorld and the DeepMind Control suite [26, 33], offline PIT-validity evaluation on real robot trajectories from DROID [35], and the financial-network instantiation in which do-queries answer contagion questions [36, 37] are the immediate next steps of this program.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The complete implementation, the ground-truth benchmark, all experiment scripts, result files, and the scripts that regenerate every table of this paper are available at `https://github.com/vinhqdang/CAIRN-A-Causally-Factored-Self-Certifying-World-Model-for-Interventional-Generalization`.

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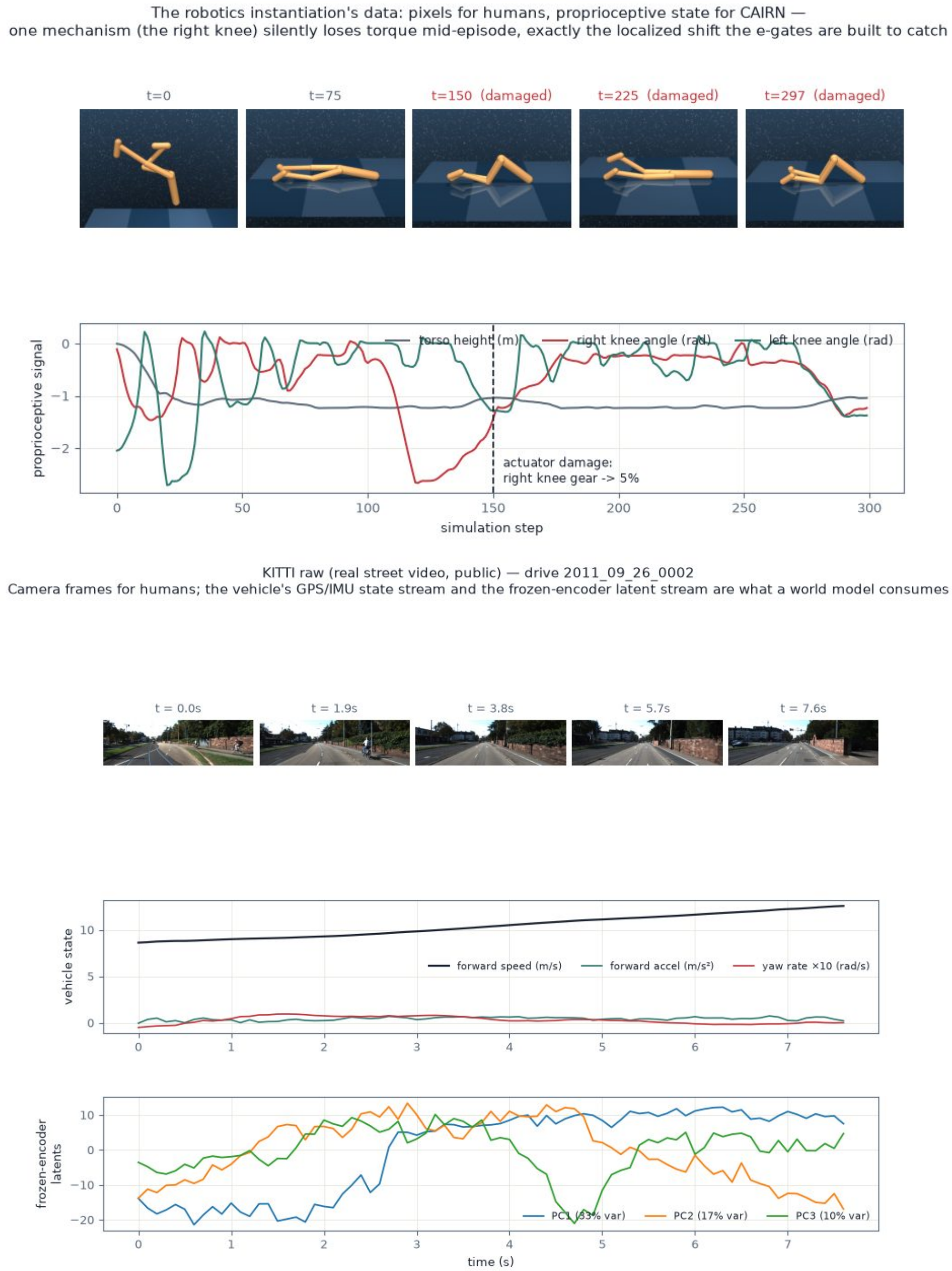

Figure 8: The road to physical instantiations, illustrated with real data processed by our pipeline. Top: a DeepMind Control walker episode in which the right-knee actuator silently loses 95% of its torque at the dashed line; one proprioceptive trace collapses while the others keep their range—a one-mechanism shift in the wild, and exactly the signature the e-gates localize. Bottom: real street video (KITTI) processed by the frozen-encoder pathway of Section 3.1: camera frames, the vehicle's measured state stream, and the first three principal components of frozen DINOv2 features per frame—a smooth latent state stream obtained from raw video without any training, over which CAIRN's mechanisms and gates operate unchanged.